

EE5311 LAB Tutorial-2 Report

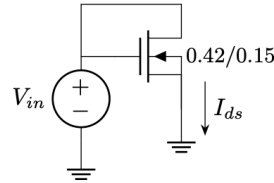
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Question :

EC 5311 Digital IC Design: Assignment 2 Transfer characteristics of transistors

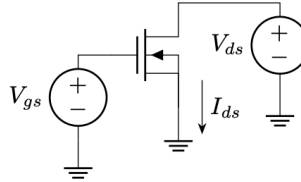
$V_{Tn} = 0.7 \text{ V},$	$\mu_n = 0.025 \text{ m}^2/\text{Vs},$	$C_{oxn} = 8.34 \text{ fF}/\mu\text{m}^2,$	$vsat_n = 8 \times 10^4 \text{ m/s},$	$\lambda_n = 0.2/\text{V}$
$ V_{Tp} = 0.7 \text{ V},$	$\mu_p = 0.009 \text{ m}^2/\text{Vs},$	$C_{oxp} = 8.16 \text{ fF}/\mu\text{m}^2,$	$vsat_p = 3 \times 10^4 \text{ m/s},$	$\lambda_p = 0.2/\text{V}$

1. (a) Consider the nMOS transistor connected as shown below.



Is it in the saturation or linear region? Using ngspice, obtain the transfer characteristics I_{ds} vs V_{gs} assuming $L = 0.15\mu\text{m}$ and $W = 0.42\mu\text{m}$. Plot the simulated model and analytical model in the same plot and find the mean percentage error.

- (b) Connect sources to the nMOS transistor as shown and obtain the output characteristics for $V_{gs} = 0.6\text{V}, 1\text{V}, 1.4\text{V}$ and 1.8V . Plot the analytical model and obtain the mean percentage error.

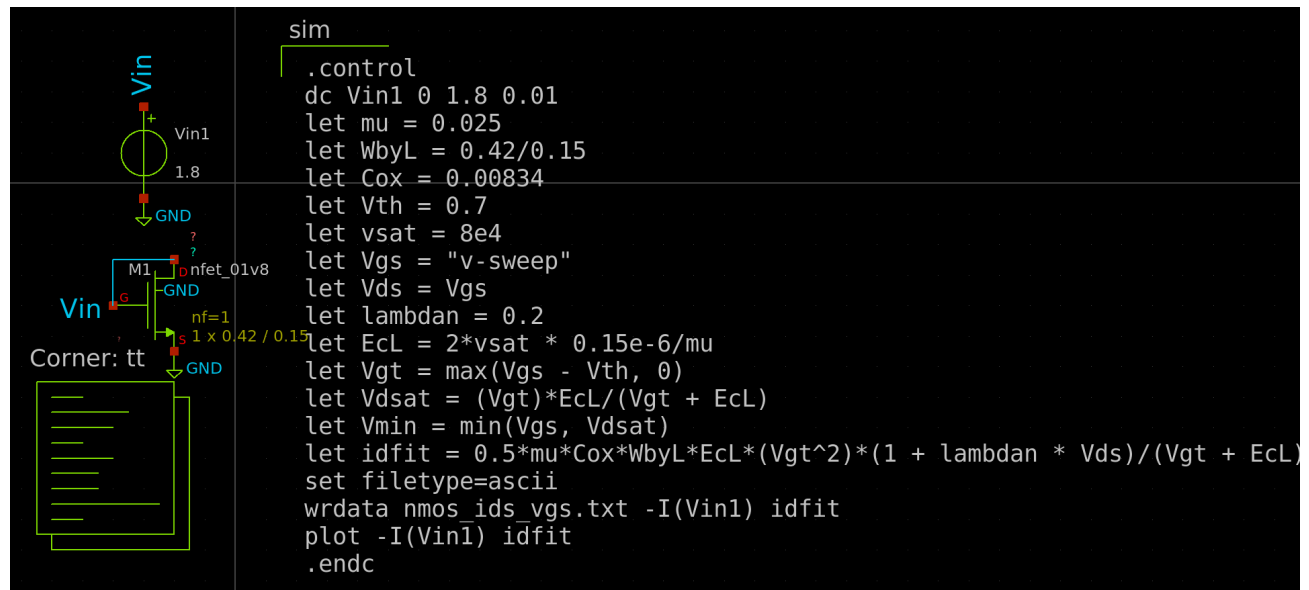


- (c) Simulate the I_{ds} vs V_{ds} for $V_{gs} = V_{DD}$ for transistors of different W/L keeping W/L ratio constant while increasing the sizes. Note how the velocity saturation effect decreases as overall length increases.

2. Repeat the exercise for the pMOS transistor.

Question 1.A:

Schematic :-



Plots :-

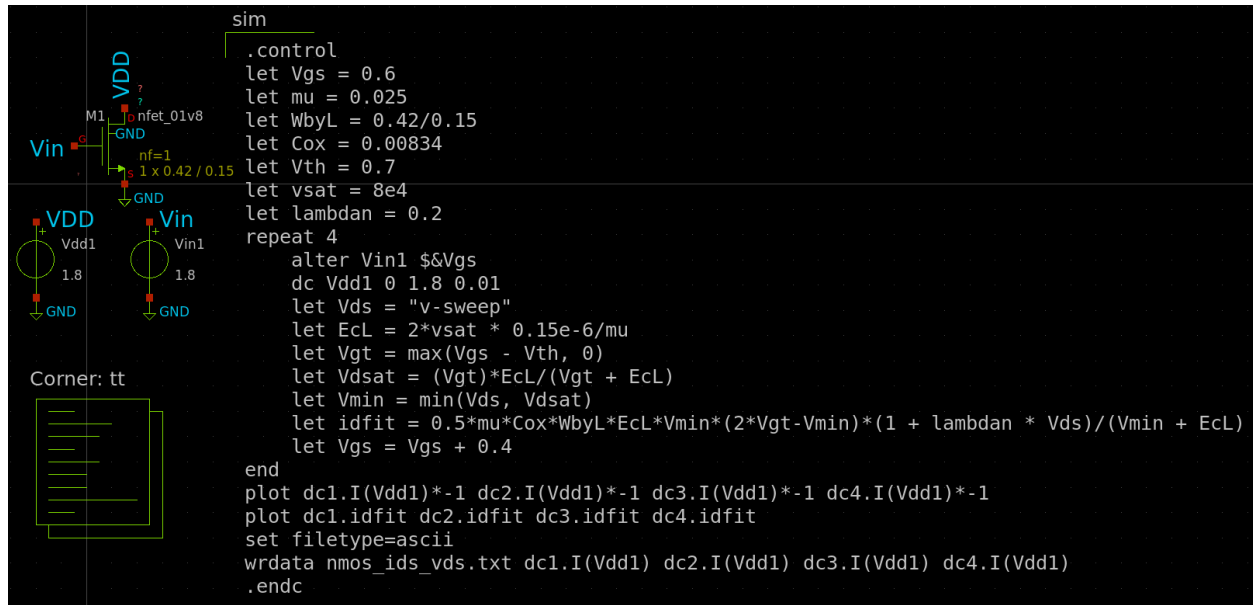
I_{ds} vs V_{gs} plot - (when $V_{gs} = V_{ds}$); Simulated - Red; Analytical - Blue



Calculated Mean Value between Simulated and Analytical =
13.36708422% (link to [NMOS Excel Sheet](#))

Question 1.B:

Schematic :-



Plots :-

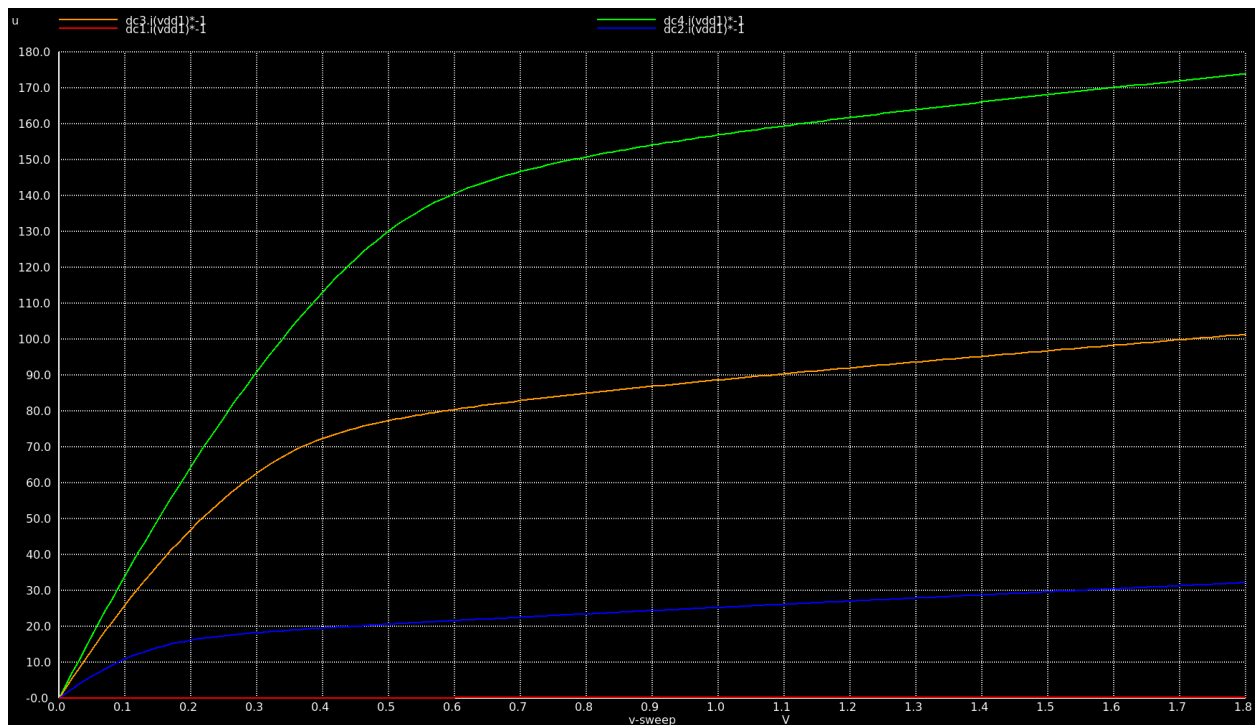
I_{ds} vs V_{gs} plot - (Simulated)

When, $V_{gs} = 0.6V$ - Red

$V_{gs} = 1V$ - Blue

$V_{gs} = 1.4V$ - Yellow

$V_{gs} = 1.8V$ - Green



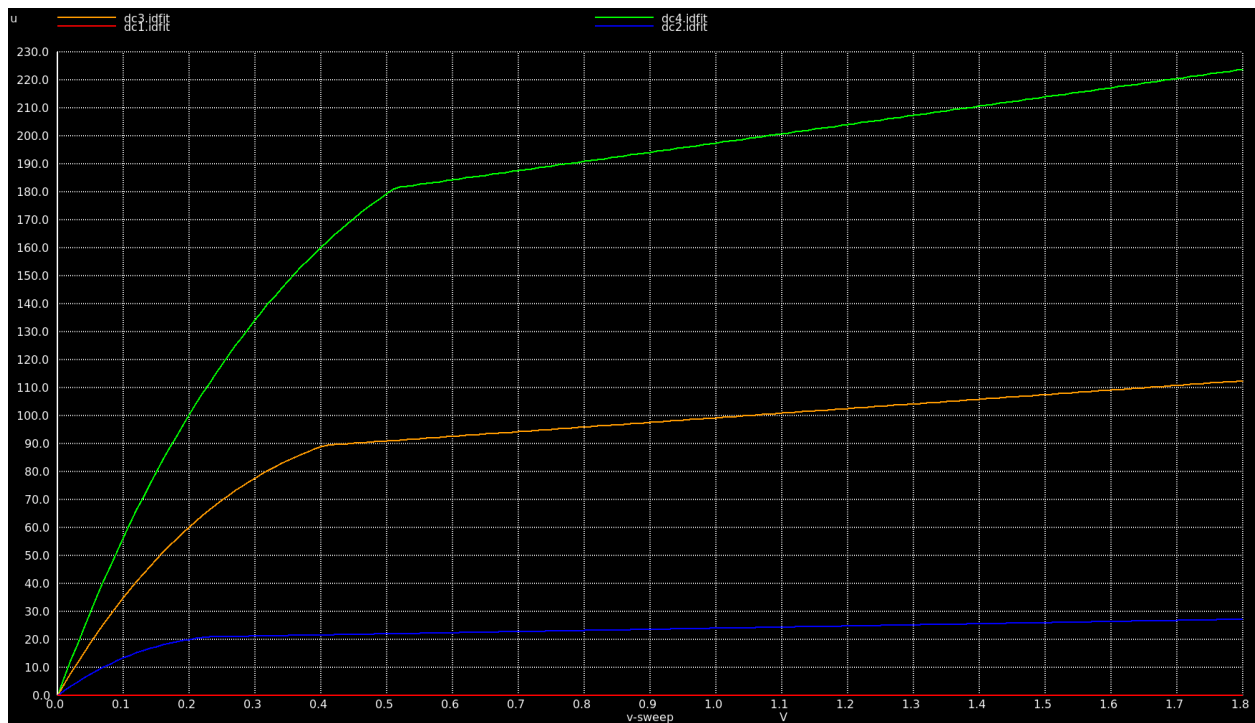
I_{ds} vs V_{gs} plot - (Analytical)

When, $V_{gs} = 0.6V$ - Red

$V_{gs} = 1V$ - Blue

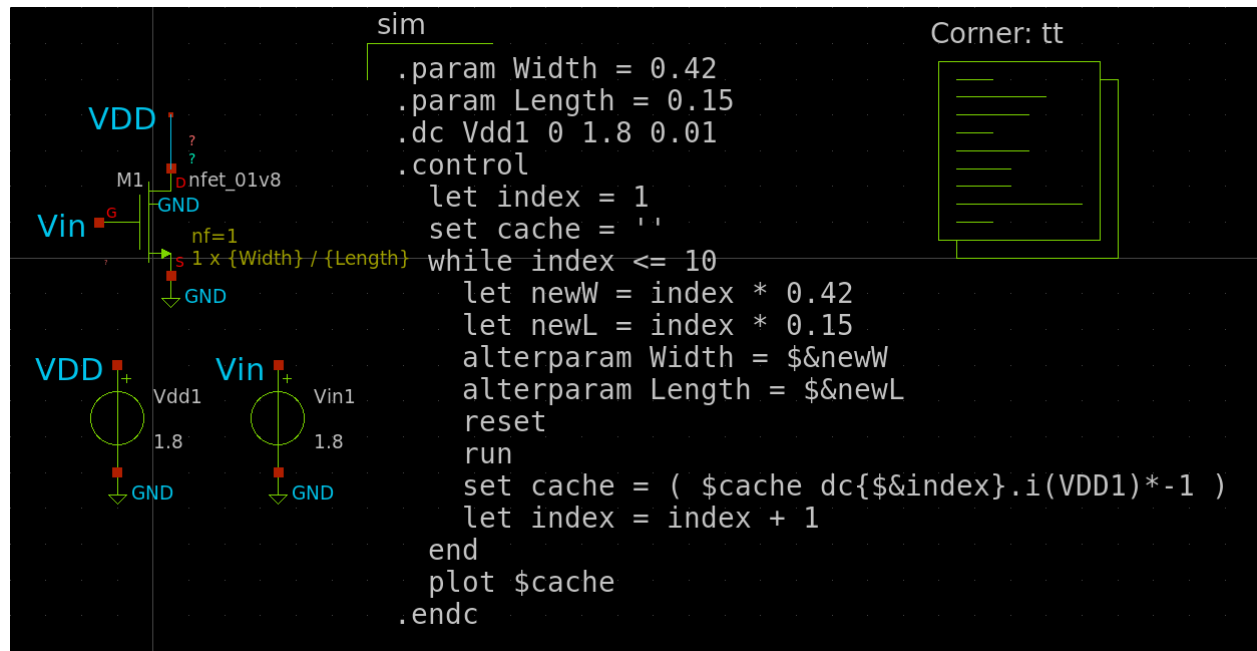
$V_{gs} = 1.4V$ - Yellow

$V_{gs} = 1.8V$ - Green



Question 1.C:

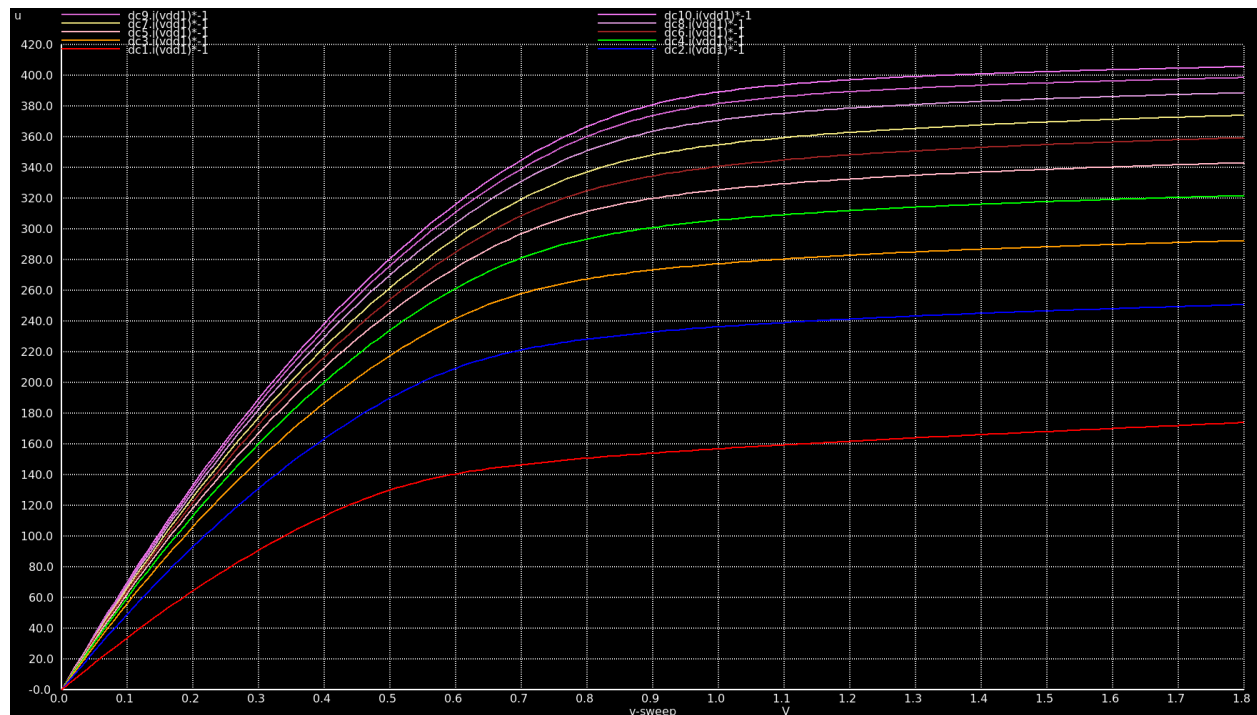
Schematic :-



Plots :-

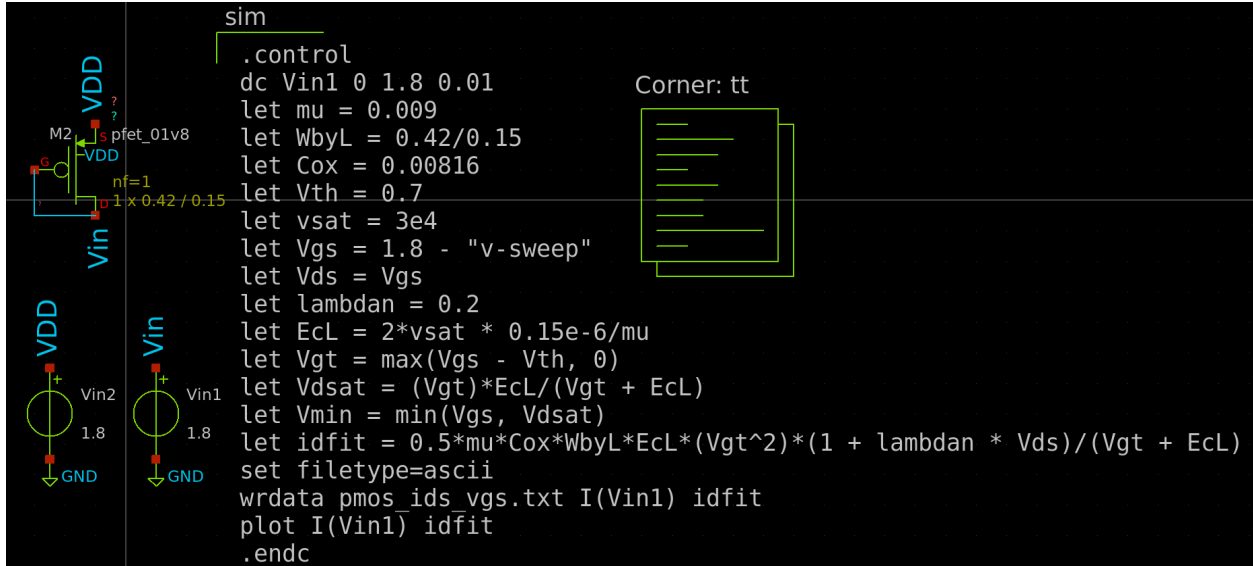
I_{ds} vs V_{gs} plot - (When $V_{gs} = 1.8V$ and W/L ratio is constant)

*but W and L values increased every time starting from $W = 0.42 \mu m$ and $L = 0.15 \mu m$ increased with same values in 10 steps



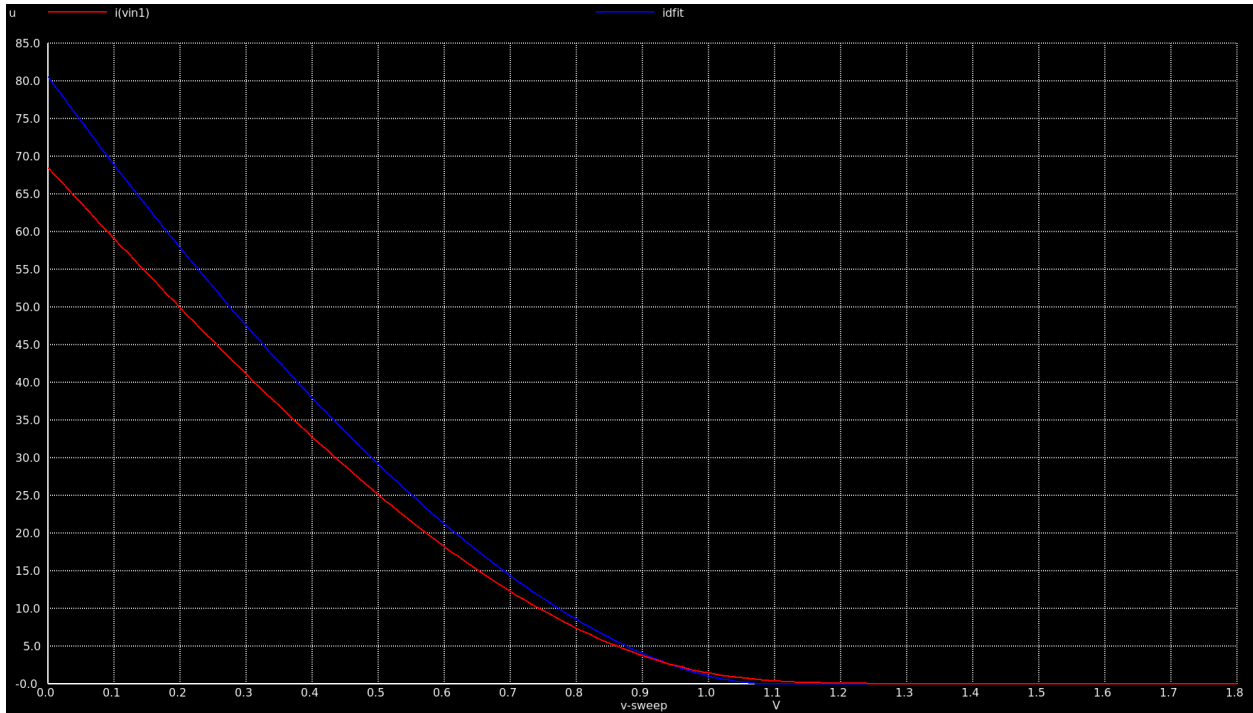
Question 2.A:

Schematic :-



Plots :-

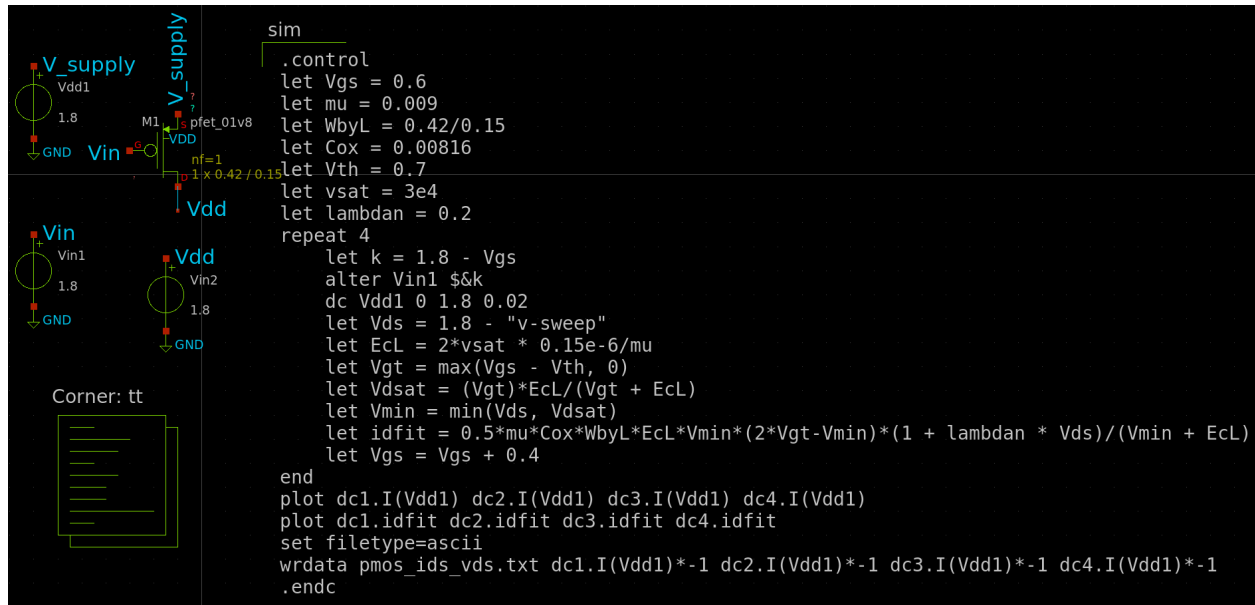
I_{ds} vs V_{gs} plot - (when $V_{gs} = V_{ds}$); Simulated - Red; Analytical - Blue



Calculated Mean Value between Simulated and Analytical =
19.35043324% (link to [PMOS Excel Sheet](#))

Question 2.B:

Schematic :-



Plots :-

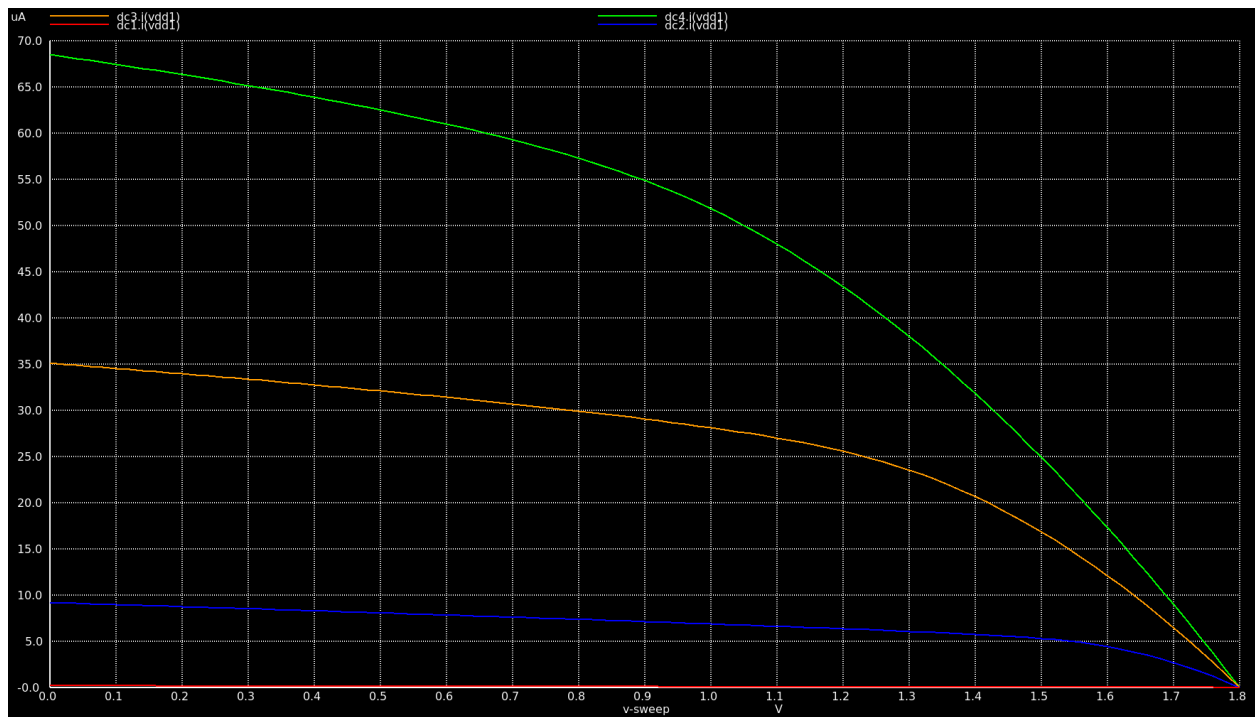
I_{ds} vs V_{gs} plot - (Simulated)

When, $V_{gs} = 0.6V$ - Red

$V_{gs} = 1V$ - Blue

$V_{gs} = 1.4V$ - Yellow

$V_{gs} = 1.8V$ - Green



I_{ds} vs V_{gs} plot - (Analytical)

When, $V_{gs} = 0.6V$ - Red

$V_{gs} = 1V$ - Blue

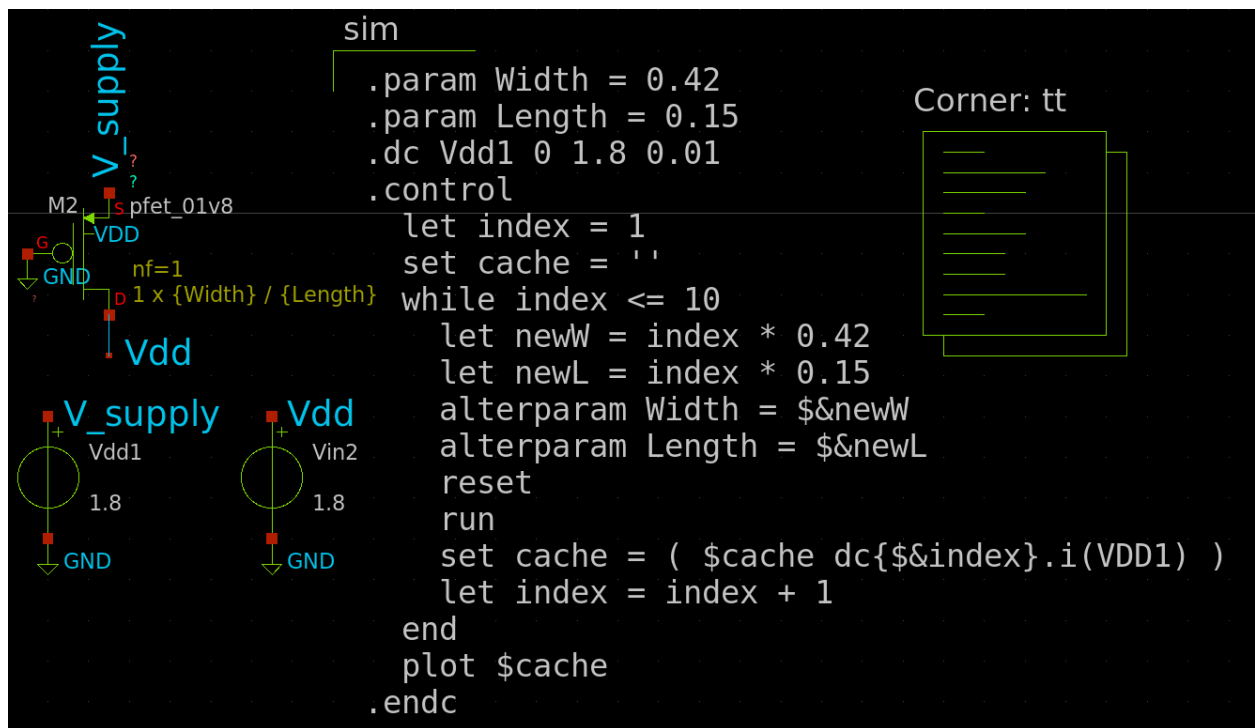
$V_{gs} = 1.4V$ - Yellow

$V_{gs} = 1.8V$ - Green



Question 2.C:

Schematic :-



Plots :-

I_{ds} vs V_{gs} plot - (When $V_{gs} = 1.8V$ and W/L ratio is constant)

*but W and L values increased every time starting from $W = 0.42 \mu m$ and $L = 0.15 \mu m$ increased with same values in 10 steps

